

Unsupervised and Semi-Supervised Anomaly Detection

on the LHC Olympics 2020 R&D Dataset

Final project report · Evidence-based reproduction and implementation audit

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Simulation-only benchmark. This report describes methods and measured performance on public simulated samples; it makes no discovery claim and does not estimate real-data physics sensitivity.

Executive summary

This project benchmarks six anomaly-detection methods on the public LHC Olympics 2020 R&D dijet samples. The verified Zenodo v5 release supplies 1.1 million 2-prong events for model development and 100,000 3-prong signal events for topology-transfer evaluation. The implementation produces deterministic event splits, validated features, model checkpoints, evaluation tables, mass diagnostics, and run metadata.

The reproduction run contains 52 unique evaluated configurations across seeds 42, 43, 44, 45, 46; 47 run records are marked completed in the run manifest. The full contamination/label-budget grid is anchored to seed 42. Headline repeated-seed summaries are reported only if the corresponding replicate table exists and includes those runs.

Main findings

Among zero-contamination, zero-labeled-signal models with seed 42, M0_TauCut has the highest held-out 2-prong ROC AUC (0.891). This is a single-seed benchmark result, not a claim of statistical superiority. Its 3-prong ROC AUC is 0.676.

Interpretation is constrained by one-seed grid results, the limited repeated-seed headline subset, simulation-only data, and the known subjettness-definition discrepancy documented below. Background mass preservation and the fixed-window null are diagnostics; the reported local significance is not a global discovery significance.

Project status

Completed: official data acquisition and checksum verification; canonical feature ingestion and low-level cross-checks; deterministic class-stratified splits; train/validation/test separation; six method implementations; contamination and label-budget grids; held-out 3-prong scoring; validation-derived operating thresholds; mass-shaping and fixed-window diagnostics; packaging, license, citation metadata, and this report. Outstanding audit limitations and deviations are listed in Section 10.

1. Research question and scope

The central question is how one-class and few-label anomaly detectors trade off detection ranking, robustness to contaminated training data, transfer to a different signal topology, and preservation of the resonant background mass spectrum. The study is a reproducible methods benchmark on simulated R&D data. It does not make a claim about discoveries, detector data, or expected experimental sensitivity.

The full project requirements are captured in PRD_final.md. The experiment uses the released 2-prong signal for development and holds the distinct 3-prong signal sample for transfer evaluation. The resonant observable mJJ is excluded from the principal detector inputs and included only in an explicit leakage-control model.

2. Dataset and provenance

The four public HDF5 files were obtained from LHC Olympics 2020 R&D dataset, Zenodo record v5 (DOI 10.5281/zenodo.6466204). The raw event files contain particle-level detector representations; released high-level feature files provide the canonical benchmark features. The data manifest records source checksums, verified SHA-256 digests, and file sizes. The raw data are not distributed with this repository.

Sample	Available rows	Use
QCD background	1,000,000	Train/validation/test background pool
2-prong signal	100,000	Development signal, contamination, labeled pool and test signal
3-prong signal	100,000	Held-out topology transfer test

Event splits are deterministic and stratified with seed 42. Invalid feature rows are excluded. The 2-prong background split contains 599,952 train, 199,984 validation, and 199,984 test rows; signal counts are 59,983, 19,994, and 19,996. The separate 3-prong sample contributes 99,975 valid held-out events.

2.1 Reproducibility inputs

The benchmark uses author-supplied high-level features as the canonical values. This avoids silently substituting a locally implemented subjettness convention for the release convention. A deterministic 128-event raw-clustering sample is compared against those supplied quantities. The optional `--from-raw` path reclusters all events and documents its exclusive-kT subjettness axes; it is not the canonical benchmark route.

3. Feature construction and validation

Principal inputs are leading-jet mass (mJ1), mass difference (dmJ), leading and subleading jet tau21, and leading-jet angular separation (dRJJ). The dijet mass mJJ is retained for mass diagnostics and the M6 leakage-control ablation. It is excluded from M1–M5 model input. Tau21 is tau2/tau1 with invalid/zero denominators identified and removed from the usable event set.

Sample	Rows	Invalid tau rows	mJ1 MAE (GeV)	dmJ MAE (GeV)	dRJJ MAE	mJJ MAE (GeV)
events_anomalydetection_v2.features.h5	1,100,000	107 (0.0097%)	0.00016	0.00025	0.00	0.0023
events_anomalydetection_Z_XY_qqq.features.h5	100,000	25 (0.0250%)	0.00023	0.00027	0.00	0.0032

Across the raw validation sample, jet kinematics and angular separation agree closely with supplied values. Locally recomputed tau21 differs materially (MAE about 0.04–0.07) because the exact released FastJet-contrib configuration is not fully encoded here. This is why the authors' supplied tau values define the primary experiment; the discrepancy is recorded rather than obscured.

4. Methods and experiment matrix

ID	Method	Role
M1	MLP autoencoder	Unsupervised reconstruction baseline
M2	Isolation Forest	Classical unsupervised baseline
M3	Deep SVDD	Unsupervised deep one-class model
M4	Deep SAD	Semi-supervised scarce-label model
M5	Supervised MLP	Fully supervised reference
M6	Mass-aware Deep SVDD	Explicit mJJ leakage control

The seed-42 grid varies signal contamination f_{train} over 0%, 0.1%, 0.5%, and 1% for M1–M3. Deep SAD varies contamination over 0%, 0.5%, and 1%, crossed with 10, 100, and 1,000 labeled signal events. M5 uses all labeled training signal available under its reference configuration; M6 tests direct mass access. Six headline configurations are additionally repeated with seeds 43 through 46 when present in the run table.

Unsupervised models use background-only training statistics. Injected unlabeled signal examples are sampled without replacement with saved event IDs. Explicitly labeled signal events are disjoint from the unlabeled pool. Feature scaling is fitted on training data only. Deep SVDD center initialization excludes the labeled anomalies in the Deep SAD path.

5. Training and evaluation protocol

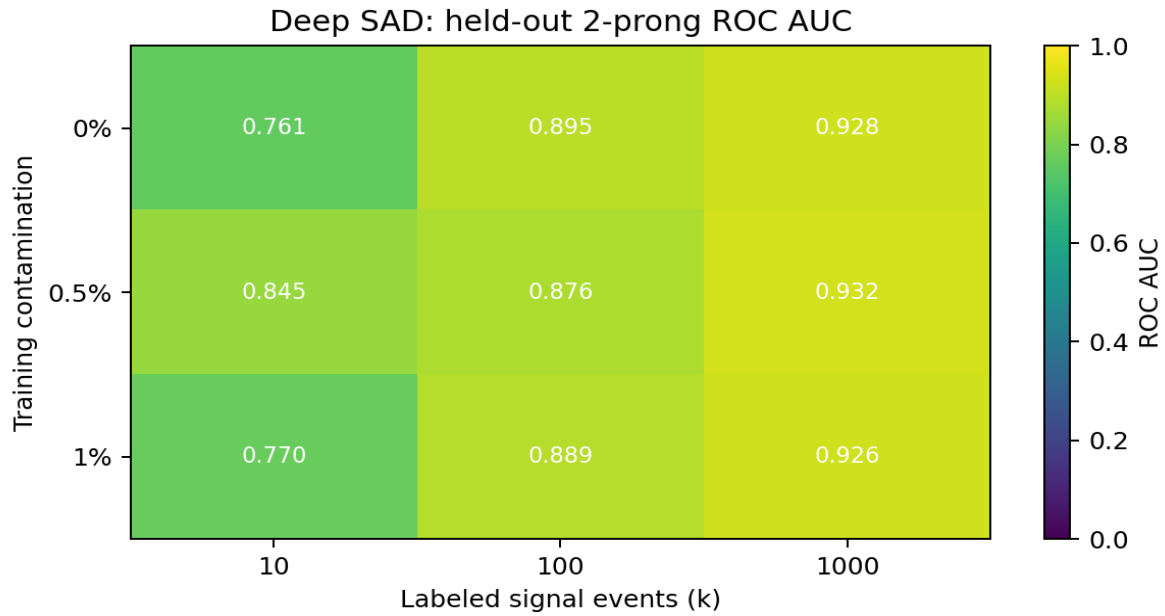
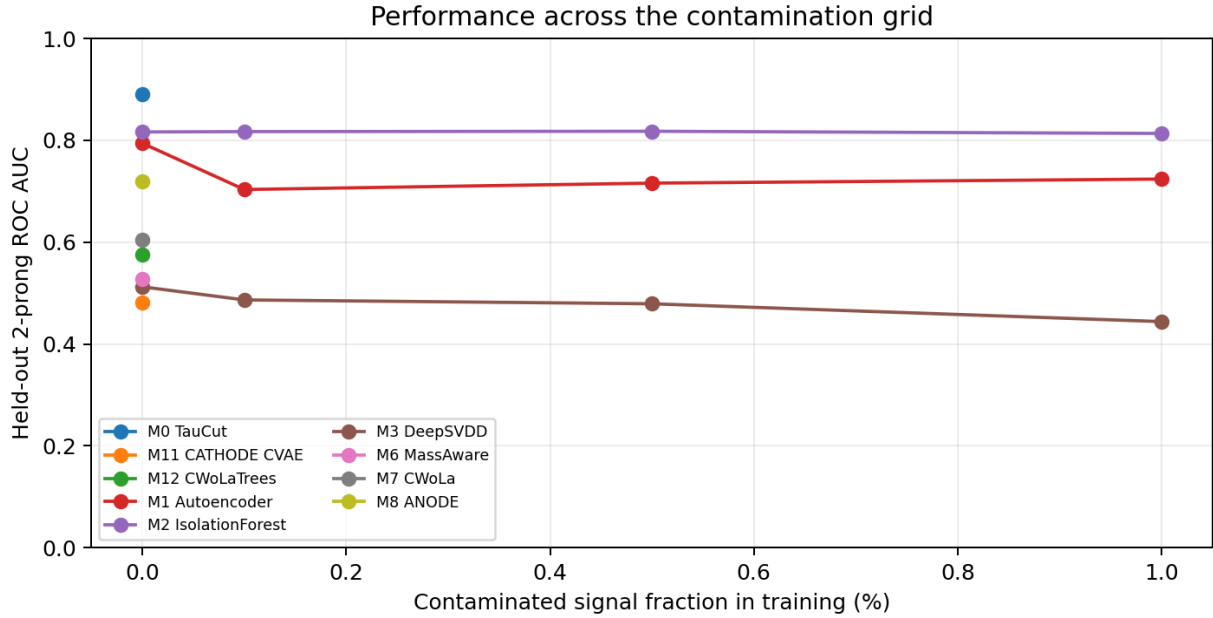
Models are trained on the training split only. Validation background scores determine the score thresholds for target background efficiencies of 10% and 1%; those thresholds are then frozen and applied to untouched test samples. Test labels are used only for final metrics. The split manifest, seeds, configuration grid, exact selected signal indices, scaler, and model checkpoint are saved alongside results.

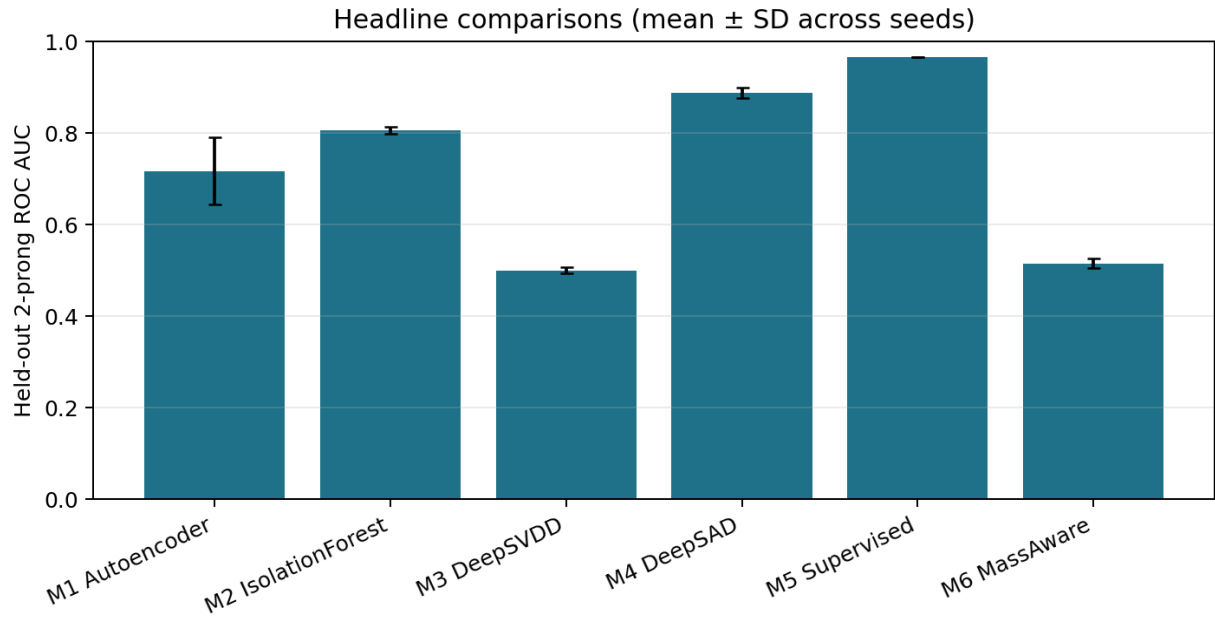
Reported ranking metrics include ROC AUC, background rejection at fixed signal efficiencies, and maximum significance-improvement characteristic. Topology transfer is measured by scoring the held-out 3-prong sample against

the 2-prong background test sample. Mass dependence is summarized by score–mJJ association, Jensen–Shannon distance between the inclusive and selected background mass distributions, and binned acceptance plots.

The fixed-window diagnostic uses the 3.3–3.7 TeV signal window with a 2.5–4.5 TeV sideband fit in 50 GeV bins. It reports local excess and fit uncertainty, plus a background-only Poisson bootstrap where available. The window is fixed; the statistic is local and must not be interpreted as a global look-elsewhere-corrected significance.

6. Measured results





Model / setting	f_train	k	AUC 2-prong	AUC 3-prong	JSD @10%	Score-mJJ
M0 TauCut	0%	0	0.891	0.676	0.003	0.062
M11 CATHODE CVAE	0%	0	0.481	0.375	0.000	-0.028
M12 CWoLaTrees	0%	0	0.575	0.616	0.189	0.451
M1 Autoencoder	0%	0	0.794	0.759	0.007	0.087
M1 Autoencoder	0.1%	0	0.703	0.670	0.011	0.044
M1 Autoencoder	0.5%	0	0.716	0.550	0.029	0.054
M1 Autoencoder	1%	0	0.724	0.644	0.009	0.047
M2 IsolationForest	0%	0	0.817	0.714	0.023	0.270
M2 IsolationForest	0.1%	0	0.817	0.716	0.023	0.269
M2 IsolationForest	0.5%	0	0.818	0.718	0.023	0.260
M2 IsolationForest	1%	0	0.814	0.716	0.023	0.262
M3 DeepSVDD	0%	0	0.512	0.636	0.001	0.022
M3 DeepSVDD	0.1%	0	0.487	0.620	0.001	0.020
M3 DeepSVDD	0.5%	0	0.479	0.619	0.001	0.019
M3 DeepSVDD	1%	0	0.444	0.618	0.001	0.013
M4 DeepSAD	0%	10	0.761	0.580	0.002	0.036
M4 DeepSAD	0%	100	0.895	0.759	0.010	0.069
M4 DeepSAD	0%	1000	0.928	0.772	0.013	0.056
M4 DeepSAD	0.5%	10	0.845	0.682	0.001	0.041
M4 DeepSAD	0.5%	100	0.876	0.650	0.002	0.031
M4 DeepSAD	0.5%	1000	0.932	0.822	0.013	0.040
M4 DeepSAD	1%	10	0.770	0.615	0.002	0.075
M4 DeepSAD	1%	100	0.889	0.706	0.004	0.041

Model / setting	f_train	k	AUC 2-prong	AUC 3-prong	JSD @10%	Score-mJJ
M4 DeepSAD	1%	1000	0.926	0.754	0.005	0.004
M5 Supervised	0%	1000	0.966	0.927	0.007	0.012
M6 MassAware	0%	0	0.528	0.683	0.026	0.067
M7 CWoLa	0%	0	0.605	0.643	0.189	0.433
M8 ANODE	0%	0	0.720	0.578	0.058	0.337

The table lists 28 seed-42 configurations. All metrics are on simulated data. AUC measures ranking and is prevalence-independent; it is not a direct estimate of discovery power. The 3-prong metric measures cross-topology ranking, not training on the 3-prong signal.

Repeated-seed summary

Model / setting	Seeds	AUC mean $\pm$ SD	3-prong AUC mean $\pm$ SD
M1_Autoencoder · f=0 · k=0	5	0.718 $\pm$ 0.074	0.597 $\pm$ 0.103
M2_IsolationForest · f=0 · k=0	5	0.807 $\pm$ 0.008	0.683 $\pm$ 0.021
M3_DeepSVDD · f=0 · k=0	5	0.500 $\pm$ 0.007	0.617 $\pm$ 0.011
M4_DeepSAD · f=0.5 · k=100	5	0.888 $\pm$ 0.011	0.749 $\pm$ 0.061
M5_Supervised · f=0 · k=1000	5	0.966 $\pm$ 0.000	0.929 $\pm$ 0.001
M6_MassAware · f=0 · k=0	5	0.515 $\pm$ 0.010	0.667 $\pm$ 0.023

7. Mass shaping and null diagnostics

For each configuration, the evaluator saves inclusive and selected mass spectra, binned background acceptance, and mean score by mass. JSD compares normalized binned distributions. Lower JSD indicates closer shapes under that binning, but does not prove absence of mass dependence or guarantee a valid search procedure. Thresholds come from validation background; test mass plots are diagnostic.

Background-only null trials repeatedly Poisson-resample the test background and refit the fixed-window sidebands. The current evaluator uses 50 pseudoexperiments per configuration. This small ensemble is a smoke-level diagnostic: its tail probability resolution is coarse (minimum 1/51), and it does not provide a calibrated global p-value. A null local significance near zero is expected when there is no injected excess.

No bump claim is made. The bootstrap and local fit are reported to characterize the diagnostic and expose fit stability. Repeated-window searches, systematic uncertainties, control-region optimization, and real detector effects are outside the present scope.

8. Reproducibility and artifacts

From a clean Python 3.11 or 3.13 environment (the final core run used Python 3.11), install the package and run `python scripts/reproduce.py`. The workflow verifies or downloads the four official data files, builds canonical feature tables, creates deterministic splits, fits the configured experiment grid, evaluates frozen thresholds, and writes the report inputs. Raw data are downloaded at runtime and excluded from version control and the Docker build context.

Key outputs include `data/manifest.yaml`, `data/splits/split_manifest.json`, feature-validation JSON files, `reports/tables/results.csv`, `reports/tables/run_manifest.json`, `replicate_summary.csv` when completed, saved experiment sample IDs, MLflow records, model checkpoints, mass diagnostic figures, and this PDF. The report is regenerated with `python scripts/build_report.py`.

The MIT license, CITATION.cff, dependency metadata, Dockerfile and provenance notes accompany the release. Reproduction requires Zenodo availability and sufficient disk space.

9. Implementation audit and limitations

The implementation audit is intentionally separated from the PRD wish list. The following boundaries remain material to interpretation:

Finding	Impact / status
No complete end-to-end three-prong training/evaluation protocol	Held-out topology is evaluated for scores; a matched 3-prong training regime is not part of the current matrix.
Released tau configuration is not reconstructed exactly	Canonical results use supplied high-level tau values; raw reclustering is validation only.
Five seeds for core headline settings; sweep mostly one seed	Small uncertainty sample; full grid robustness is not seed-averaged.
50-trial null bootstrap	Tail resolution and calibration are limited; no global significance claim.
Fixed-window diagnostic and simplified sideband fit	Not a complete bump-hunt search procedure or systematic model.
Simulation-only R&D samples	No detector-data or experimental-sensitivity claim.
Author clean-checkout reproduction completed	All 47 core fits rerun in an isolated Python 3.11 environment; no external independent review.

The prior audit's concrete fixes are reflected in the current sources: no dummy results are treated as evidence; subjectivity/tau features are computed or sourced explicitly rather than set to constants; train/validation/test roles are separated; thresholds are selected from validation; signal contamination is sampled with saved event IDs; null trials and mass diagnostics are emitted; and repository license/citation/packaging files are populated. Where experiments remain incomplete, the report states that directly.

Release verification

All 47 core configurations were trained from scratch in a fresh checkout and isolated Python 3.11 environment, using checksum-verified raw inputs. This fresh run supplies the released core tables; older measurements are preserved separately. The six headline settings have five seeds each. Finite-test AUC uncertainty uses 500 paired stratified bootstrap trials on frozen seed-42 scores, reported separately in reports/tables/auc_bootstrap.json. Other metrics do not have finite-test confidence intervals. Runtime, disk use and differences from historical results are recorded in reports/reproduction/verification.json.

10. Conclusions and next steps

The project provides a provenance-tracked benchmark for ranking performance, contamination response, scarce-label learning, topology transfer and mass shaping. Its simulation-only results do not establish a universal winner.

Further scientific work should repeat the full grid across seeds, enlarge the predeclared null ensemble, add detector systematics and seek independent review of the released tau definition.

11. References

- [1] LHC Olympics 2020 R&D dataset, Zenodo record 6466204, version 5. <https://doi.org/10.5281/zenodo.6466204>
- [2] G. Kasieczka et al., "The LHC Olympics 2020: A Community Challenge for Anomaly Detection in High Energy Physics," arXiv:2101.08320 (2021). <https://arxiv.org/abs/2101.08320>
- [3] L. Ruff et al., "Deep One-Class Classification," ICML 2018. <https://proceedings.mlr.press/v80/ruff18a.html>
- [4] L. Ruff et al., "Deep Semi-Supervised Anomaly Detection," ICLR 2020. <https://openreview.net/pdf?id=HkgH0TEYwH>
- [5] Project protocol and requirements: PRD_final.md in this repository.